06

Lecture 06

E-UTRAN: the LTE Radio Access Network

E-UTRAN: شبکهٔ دسترسی رادیویی LTE

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
7–9 hours (3.5 h reading, 4–5 h exercises)

PREREQUISITES
Lecture 5; the OSI layering; basic ARQ

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-06-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the functions of the eNB and explain why the flat E-UTRAN could absorb the RNC.
- **LO2** **Draw** the LTE control-plane and user-plane protocol stacks and state the role of PHY, MAC, RLC, PDCP and RRC.
- **LO3** **Map** logical channels to transport and physical channels in both directions.
- **LO4** **Explain** the S1 and X2 interfaces, their protocol stacks and main procedures.
- **LO5** **Describe** the RRC state machine, DRX and the NR RRC_INACTIVE extension, and **quantify** the power/latency trade-off of DRX.
- **LO6** **Size** PDCP sequence numbers and RLC windows for a target data rate.

2 Prerequisite knowledge

This lecture zooms into the left half of the EPS architecture of Lecture 5. You need the idea of protocol layering (each layer offers a service to the one above and adds a header), the notion of automatic repeat request (ARQ) with sequence numbers and windows, and the EPS bearer concept.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
E-UTRAN	شبکهٔ دسترسی رادیویی LTE	The LTE RAN: a mesh of eNBs, no controller.
RRC	کنترل منابع رادیویی	Layer-3 control between UE and eNB: connection, measurements, security.
PDCP	پروتکل همگرایی دادهٔ بسته‌ای	Header compression, ciphering, integrity, in-order delivery.
RLC	کنترل پیوند رادیویی	Segmentation and ARQ in TM, UM or AM mode.
MAC	کنترل دسترسی به رسانه	Multiplexing, scheduling, HARQ, random access.
Logical channel	کانال منطقی	*What* is carried (control or traffic).
Transport channel	کانال انتقال	*How* it is carried (format, coding) between MAC and PHY.
Physical channel	کانال فیزیکی	The actual time–frequency resources.
S1AP / X2AP	پروتکل‌های کاربردی S1 و X2	Signalling eNB–MME and eNB–eNB over SCTP.
DRX	دریافت ناپیوسته	Sleep cycles that save battery at the cost of latency.

3 Introduction and motivation

In UMTS the radio protocols were split across two nodes: the NodeB handled the physical layer, the RNC everything above it. Every scheduling decision, every retransmission of RLC and every handover therefore crossed the Iub transport network, adding delay and making the RNC a bottleneck. HSPA already moved fast scheduling and HARQ to the NodeB. LTE completed the move: the eNB terminates *all* radio protocols, so the round trip between a radio event and a decision stays inside one box.

This lecture opens that box. We follow user data and signalling down the protocol stack, see how the different kinds of information are organised into channels, and study the two network interfaces of the eNB: S1 towards the core and X2 towards its neighbours. We end with the RRC state machine, which decides how much the UE and the network must keep in memory and how fast an idle phone can start sending data.

4 Fundamental concepts

4.1 Why no RNC?

The RNC existed for three reasons: soft-handover combining, central radio resource management of many NodeBs, and termination of protocols that were too complex for 1990s base stations. LTE removed each reason: OFDMA uses hard handover, so no combining is needed; scheduling every 1 ms must be done next to the radio; and processing power had become cheap. Inter-cell coordination that the RNC used to provide is replaced by direct eNB-to-eNB signalling over X2 and by self-organising network functions.

4.2 Layering and the bearer service

Each layer of the radio stack offers a well-defined service upwards:

- **PHY** offers *transport channels* to MAC.
- **MAC** offers *logical channels* to RLC.
- **RLC** offers *RLC channels* to PDCP.
- **PDCP** offers *radio bearers* to IP (data radio bearers, DRBs) and to RRC (signalling radio bearers, SRBs).

A data radio bearer is the radio segment of an EPS bearer (Lecture 5); SRB0 carries RRC messages before security is active, SRB1 and SRB2 carry RRC and NAS messages afterwards.

5 Detailed technical explanation

5.1 eNB functions

The eNB performs [1]:

- **Radio resource management**: radio bearer control, radio admission control, connection mobility control (handover), dynamic allocation of resources to UEs in uplink and downlink (scheduling), inter-cell interference coordination (ICIC), load balancing.
- **Header compression and encryption** of user data (PDCP).
- **MME selection** at attach (NAS node selection function) and routing of user data towards the S-GW.
- **Paging** and **system information broadcast**, originating from the MME and O&M respectively.
- **Measurement configuration** for mobility and scheduling.

5.2 The protocol layers

Physical layer (PHY). OFDMA in the downlink and SC-FDMA in the uplink with 15 kHz subcarriers; turbo coding for data; QPSK, 16QAM and 64QAM (256QAM in later releases); up to 4×4 (later 8×8) MIMO; HARQ soft combining; cell search with primary and secondary synchronisation signals (PSS/SSS). The resource grid and its numerology are the subject of Lecture 7.

MAC. Maps logical channels to transport channels, multiplexes several logical channels into one transport block, performs the *scheduling* decision every TTI, runs HARQ (8 parallel stop-and-wait processes in FDD), handles logical-channel priorities, random access (Lecture 8), timing advance and buffer status and power headroom reports [2].

RLC. Has three modes [4]:

- **Transparent mode (TM)**: no header, no segmentation – used for broadcast, paging and SRB0.
- **Unacknowledged mode (UM)**: segmentation/concatenation and reordering, no retransmission – used for delay-sensitive traffic such as VoLTE.
- **Acknowledged mode (AM)**: adds ARQ with status reports – used for TCP traffic and signalling, where reliability matters more than delay.

RLC adapts IP packets to the transport-block size chosen by the scheduler: it may cut a packet into segments or pack several into one PDU.

PDCP. [3] performs robust header compression (ROHC) of IP/UDP/RTP headers, ciphering of user and control data, integrity protection of control data (SRBs), and in-order, duplicate-free delivery across handover (using the SN status transfer of Lecture 4). Each PDCP PDU carries a sequence number (SN); together with a hyper-frame number (HFN) kept at both ends it forms the 32-bit *COUNT*, an input to the ciphering algorithm that must never repeat under the same key.

RRC. [5] broadcasts system information (MIB on PBCH, SIBs on PDSCH), manages the RRC connection (setup, reconfiguration, re-establishment, release), configures measurements and receives reports, activates access-stratum security, and transfers NAS messages.

5.3 Channels

Figure 1 shows the main mapping. The logical channels are the broadcast (BCCH), paging (PCCH), common control (CCCH, before an RRC connection exists), dedicated control (DCCH) and dedicated traffic (DTCH) channels. Transport channels group them by transmission format: the downlink and uplink shared channels (DL-SCH, UL-SCH) carry almost everything. Physical channels include the shared data channels PDSCH and PUSCH, control channels without transport channels (PDCCH for scheduling assignments, PCFICH for the size of the control region, PHICH for uplink HARQ acknowledgements, PUCCH for uplink control) and reference signals (CRS, DMRS, SRS) used for measurements and demodulation.

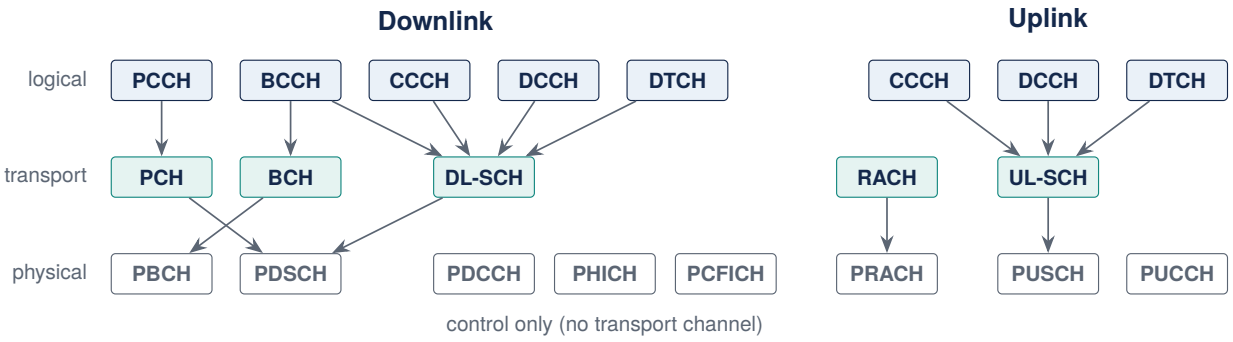


Figure 1. Main LTE channel mapping. Almost all logical channels converge on the shared channels; PDCCH, PHICH, PCFICH and PUCCH carry layer-1/2 control without a transport channel.

5.4 The S1 interface

S1 connects each eNB to one or more MMEs (S1-MME) and S-GWs (S1-U) – a many-to-many relation called *S1-flex* that allows MME pooling (Lecture 12).

- **S1-MME**: S1AP over SCTP over IP [6]. SCTP provides reliable, message-oriented transport with multi-homing and multiple streams, avoiding head-of-line blocking between UEs.
- **S1-U**: GTP-U over UDP over IP; one tunnel per E-RAB (the S1 part of an EPS bearer).

Main S1AP procedures: *S1 Setup* (eNB and MME exchange identities and served tracking areas), *Initial UE Message* (first NAS message of a UE), *Initial Context Setup* (MME installs the UE context: security key, bearers, capabilities), *E-RAB Setup/Modify/Release*, *Paging*, *Handover Preparation/Resource Allocation* (S1 handover), *Path Switch*, *UE Context Release* and *Downlink/Uplink NAS Transport*.

5.5 The X2 interface

X2 connects neighbouring eNBs [7]: X2AP over SCTP for control, GTP-U for forwarding user data during handover. Main procedures: *X2 Setup* (exchange of served cells, enabling automatic neighbour relations), *Handover Preparation*, *SN Status Transfer*, *UE Context Release*, *Load Indication* (interference information for ICIC) and *Resource Status Reporting* (load for mobility load balancing). X2 is optional: without it, handovers go through S1 and the MME.

5.6 RRC states and DRX

LTE has two RRC states (Figure 2):

- **RRC_IDLE**: no RRC connection, no context in the eNB; the UE performs cell reselection, reads system information and monitors paging once per paging cycle (idle DRX); the MME knows it at tracking-area level (ECM-IDLE).
- **RRC_CONNECTED**: the eNB holds the UE context; the UE can be scheduled, reports measurements and is handed over by the network; to save power it may use connected-mode DRX, waking for an *on-duration* once per DRX cycle.

Moving from IDLE to CONNECTED requires random access, RRC connection setup, a NAS service request, security activation and bearer set-up – tens of milliseconds and a dozen messages. NR therefore adds **RRC_INACTIVE** [8]: the UE and the gNB keep the context (security, bearers), the core still sees the UE as connected, and resumption needs only a short exchange.

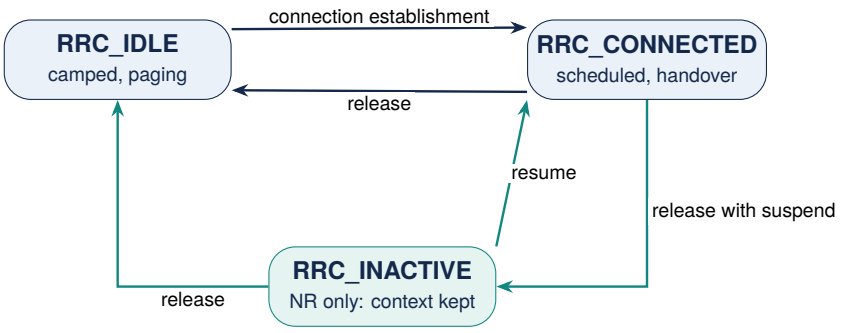


Figure 2. RRC state machine. LTE has IDLE and CONNECTED; NR adds INACTIVE, in which the context is suspended rather than released.

6 Mathematical models

6.1 Header compression gain

Problem. How much radio capacity does ROHC save for VoLTE?

Model. A speech frame of L_p bytes is carried with H bytes of RTP/UDP/IP headers; ROHC reduces H to a compressed header of H_c bytes in steady state. The radio efficiency of the packet is

$$\eta = \frac{L_p}{L_p + H}, \qquad G = \frac{L_p + H}{L_p + H_c} \text{ (capacity gain)}. \tag{1}$$

Example. $L_p = 32$ bytes, $H = 40$ bytes (IPv4 20 + UDP 8 + RTP 12), $H_c = 3$ bytes: η rises from 44% to 91% and $G = 72/35 = 2.06$. With IPv6 ($H = 60$) the gain is $92/35 = 2.6$. ROHC roughly doubles the number of VoLTE users a cell can carry; it is not applied on S1-U, where the headers are carried in full (Lecture 5).

Limits. The steady-state compressed size is reached only after a context is established; packet loss can desynchronise the decompressor, so ROHC profiles trade compression against robustness.

6.2 Sequence-number space and wrap-around

Problem. How many PDCP SN bits are needed at a given rate?

Model. A window-based protocol with an n-bit SN can have at most 2^{n-1} PDUs outstanding (half the space), otherwise old and new PDUs become ambiguous. With a PDU rate λ_p (PDUs per second), the window covers the time

$$T_{\text{win}} = \frac{2^{n-1}}{\lambda_p}, \qquad \lambda_p = \frac{R}{8L}, \tag{2}$$

for a bit rate R and PDUs of L bytes. T_{win} must exceed the longest reordering or retransmission delay, typically including HARQ, RLC ARQ and handover data forwarding.

Example. $R = 300$ Mbit/s, $L = 1500$ bytes: $\lambda_p = 25\,000$ PDU/s. A 12-bit SN gives $T_{\text{win}} = 2048/25\,000 = 82$ ms – uncomfortably short during handover. This is why LTE added a 15-bit (655 ms) and NR an 18-bit PDCP SN.

6.3 RLC acknowledged-mode throughput

For a window of W PDUs of L bytes and a status round-trip time T_{RTT}, a window-limited sender cannot exceed

$$R_{\max} = \frac{8\,W\,L}{T_{\text{RTT}}}. \tag{3}$$

With the LTE RLC AM 10-bit SN, $W = 512$; for $L = 1500$ bytes and $T_{\text{RTT}} = 20$ ms, $R_{\max} = 512 \times 12\,000/0.02 = 307$ Mbit/s: enough for Release 8, the reason for longer SNs in later releases with higher rates (Exercise 3).

6.4 DRX: power versus latency

Problem. How long should a connected UE sleep?

Model. DRX cycle T_{DRX} with on-duration T_{on}; power P_{on} when awake and P_{sleep} when asleep; downlink packets arrive at random instants (no traffic during the sleep period otherwise).

Derivation. With duty cycle $d = T_{\mathrm{on}}/T_{\mathrm{DRX}}$,

$$\bar{P} = d\,P_{\mathrm{on}} + (1-d)\,P_{\mathrm{sleep}}, \qquad \bar{D}_{\mathrm{DRX}} \approx \frac{T_{\mathrm{DRX}}}{2}, \tag{4}$$

because a packet arriving at a uniformly random time waits on average half a cycle for the next on-duration.

Example and interpretation. $T_{\mathrm{on}} = 10\,\mathrm{ms}$, $P_{\mathrm{on}} = 500\,\mathrm{mW}$, $P_{\mathrm{sleep}} = 10\,\mathrm{mW}$ (illustrative): a 40 ms cycle gives 132.5 mW and 20 ms extra delay; a 320 ms cycle 25.3 mW and 160 ms; 2560 ms gives 11.9 mW but 1.28 s (Figure 3). Power falls like $1/T_{\mathrm{DRX}}$ until P_{sleep} dominates, while latency grows linearly: beyond a few hundred milliseconds longer cycles buy little energy at a large latency cost. Real devices also stay awake for an inactivity timer after each packet, which the model ignores.

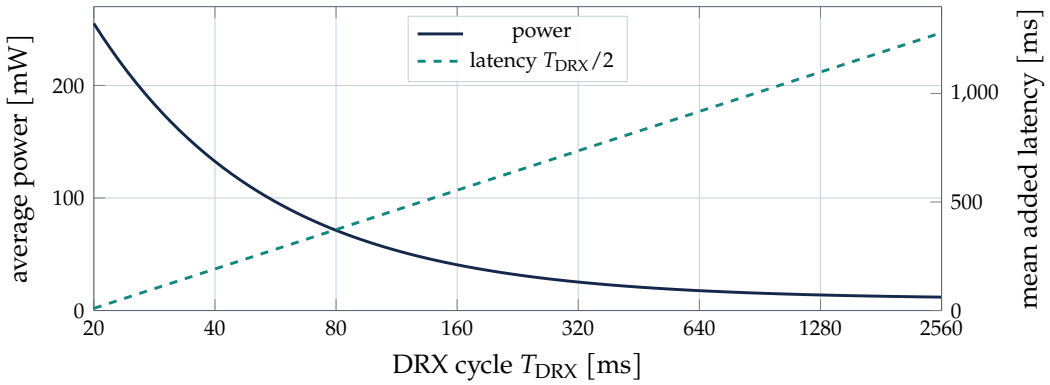


Figure 3. Average power (left axis) and mean added latency (right axis) versus DRX cycle for $T_{\mathrm{on}} = 10\,\mathrm{ms}$, from Eq. (4).

7 Network architecture and diagrams

Figure 4 summarises the radio stacks. On the control plane, RRC terminates in the eNB while NAS passes through it transparently to the MME (carried in RRC towards the UE and in S1AP towards the MME). On the user plane, PDCP, RLC and MAC terminate in the eNB, which relays the IP packets into GTP-U tunnels on S1-U.

8 Step-by-step procedures

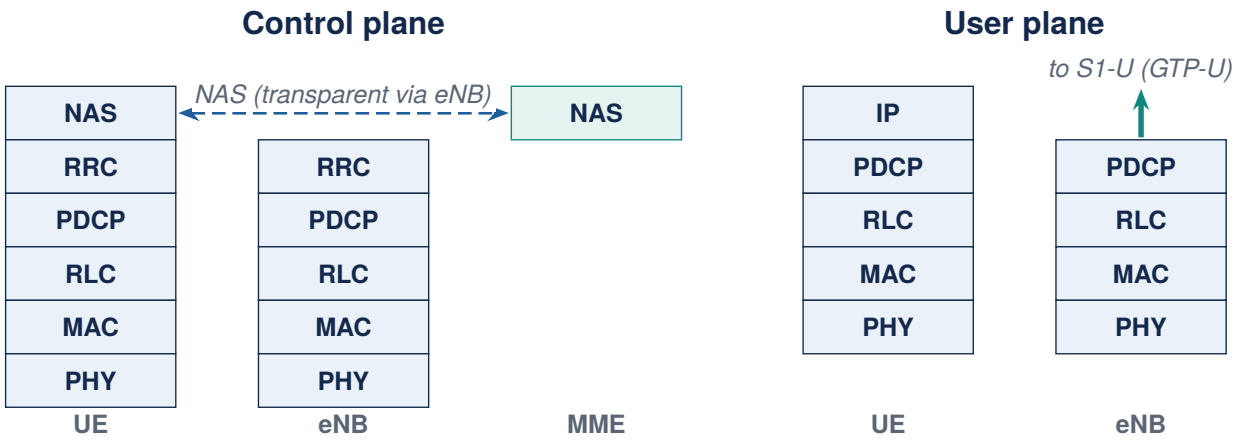


Figure 4. LTE radio protocol stacks: control plane (left) and user plane (right).

Procedure 1 • From RRC_IDLE to data transfer (mobile-originated)

1. **Random access.** The UE sends a preamble, receives a random access response with a timing advance and grant (Lecture 8).
2. **RRC connection request.** On SRB0 it sends RRC Connection Request (identity and cause); the eNB answers RRC Connection Setup, creating SRB1.
3. **NAS transfer.** The UE sends RRC Connection Setup Complete containing the NAS Service Request; the eNB forwards it in S1AP Initial UE Message to the MME selected from the UE's identity.
4. **Context set-up.** The MME answers S1AP Initial Context Setup Request with the security key K_{eNB}, the E-RABs and the S-GW tunnel endpoints.
5. **AS security.** The eNB sends Security Mode Command; the UE derives the AS keys and answers Security Mode Complete. PDCP now ciphers and integrity-protects.
6. **Bearer configuration.** The eNB sends RRC Connection Reconfiguration creating SRB2 and the DRBs; after Reconfiguration Complete it answers Initial Context Setup Response with its own S1-U tunnel endpoints. Uplink and downlink data flow.

9 Worked examples

Worked example 1 • Protocol overhead of a TCP segment

Problem. A 1460-byte TCP payload is sent over LTE with IPv4. Count the header bytes added on the radio by PDCP (2 bytes, 12-bit SN), RLC AM (2 bytes, no segmentation) and MAC (2 bytes subheader), and on S1-U.
Solution. IP packet: $1460 + 20 + 20 = 1500$ bytes. Radio: $1500 + 2 + 2 + 2 = 1506$ bytes (0.4% overhead; ROHC is usually not used for TCP). S1-U: $1500 + 36 = 1536$ bytes (2.4%).
Interpretation. For large packets layer-2 overhead is negligible; the real costs are the reference signals and control channels of the PHY (Lecture 5) and the retransmissions.

Worked example 2 • PDCP window during handover

Problem. A UE receives 150 Mbit/s in 1500-byte packets. During handover, forwarded packets may arrive up to 60 ms late. Is a 12-bit PDCP SN sufficient?
Solution. $\lambda_p = 150 \times 10^6 / 12\,000 = 12\,500$ PDU/s. Equation (2): $T_{win} = 2048/12\,500 = 164$ ms > 60 ms. Yes, with margin; at 300 Mbit/s (82 ms) the margin would be thin.

Worked example 3 • Choosing a DRX cycle

Problem. A messaging application tolerates 200 ms extra latency on average. Using the parameters of Section 6.4, choose the longest standard DRX cycle among 40, 80, 160, 320 ms and compute the power saving relative to no DRX (P_{on} continuously).
Solution. $T_{DRX}/2 \leq 200$ ms allows 320 ms (160 ms latency). $\bar{P} = 25.3$ mW versus 500 mW: a 95% reduction.
Interpretation. Most of the saving is already obtained at 160 ms (40.6 mW); the operator can choose per application through the inactivity timer and cycle settings.

10 Technology comparison

Table 1. Radio access network functions: UTRAN versus E-UTRAN versus NG-RAN.

	UTRAN (3G)	E-UTRAN (4G)	NG-RAN (5G)
Nodes	NodeB + RNC	eNB	gNB (optionally CU + DU)
RLC/PDCP location	RNC	eNB	DU (RLC), CU (PDCP)
Scheduler	RNC (R99), NodeB (HSPA)	eNB, 1 ms	gNB-DU, slot/mini-slot
Inter-node link	Iur	X2	Xn; F1 between CU and DU
Core interface	Iu (RANAP)	S1 (S1AP)	NG (NGAP)
RRC states	Idle, Cell_DCH/FACH/PCH, URA_PCH	Idle, Connected	Idle, Inactive, Connected
New L2 sublayer	–	–	SDAP (QoS-flow mapping)

Trade-off. Putting everything in the eNB minimised latency but made coordination between cells (interference, joint transmission) harder, since it now requires X2 signalling between peers. 5G partially re-centralises the RAN with the CU/DU split (Lecture 14): the time-critical lower layers stay at the antenna site, PDCP and RRC can be pooled – a deliberate compromise between UTRAN's centralisation and E-UTRAN's flatness.

11 Key takeaways

Key takeaways

- The eNB terminates PHY, MAC, RLC, PDCP and RRC: radio decisions stay in one node.
- PDCP: ROHC, ciphering, integrity (SRBs), in-order delivery; COUNT = HFN ‖ SN must never repeat.
- RLC: TM, UM (voice), AM (TCP, signalling); MAC: multiplexing, scheduling, HARQ.
- Logical → transport → physical channels; almost everything uses the shared channels.
- S1-MME (S1AP/SCTP) and S1-U (GTP-U/UDP) towards the core; X2 between eNBs.
- RRC_IDLE vs RRC_CONNECTED (+ RRC_INACTIVE in NR); DRX trades power ($\propto 1/T_{\mathrm{DRX}}$) for latency ($T_{\mathrm{DRX}}/2$).
- SN space must cover the reordering time: $2^{n-1}/\lambda_{\mathrm{p}} > T_{\mathrm{reorder}}$.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which layer?

For each function name the LTE layer that performs it: HARQ, ARQ, ciphering of user data, segmentation, header compression, integrity protection of RRC messages, choice of MCS, measurement configuration.

Exercise 2 • Level 1 – conceptual • Why SCTP?

Explain why S1AP and X2AP use SCTP rather than TCP or UDP. Give one failure scenario that multi-homing handles.

Exercise 3 • Level 2 – numerical • RLC window

Using Eq. (3), find the RLC AM window needed for 1 Gbit/s with 1500-byte PDUs and a 15 ms status round trip, and the minimum number of SN bits.

Exercise 4 • Level 2 – numerical • ROHC and IPv6

Repeat Section 6.1 for AMR-WB at 23.85 kbit/s (61-byte frames every 20 ms) with IPv6. What is the bit rate on the radio with and without ROHC, and the capacity gain?

Exercise 5 • Level 3 – analytical • DRX with an inactivity timer

After each received packet the UE stays awake for an inactivity timer T_{in}. Packets arrive as a Poisson process with rate λ. Derive an approximate expression for the fraction of time awake, assuming $T_{\text{in}} \ll 1/\lambda$, and evaluate it for $\lambda = 0.2\,\text{s}^{-1}$, $T_{\text{in}} = 100\,\text{ms}$, $T_{\text{on}} = 10\,\text{ms}$, $T_{\text{DRX}} = 320\,\text{ms}$.

Exercise 6 • Level 3 – analytical • Idle versus connected

Releasing a UE to RRC_IDLE after inactivity saves eNB memory and UE power, but each new data burst then costs about 12 signalling messages. If the eNB can handle 10^4 such transitions per second and each UE generates a burst every 30 s, how many UEs can be served purely from the signalling point of view? Discuss how RRC_INACTIVE changes the answer.

Solutions

Solution to Exercise 1

HARQ: MAC. ARQ: RLC (AM). Ciphering of user data: PDCP. Segmentation: RLC. Header compression: PDCP. Integrity of RRC: PDCP. Choice of MCS: MAC scheduler (with link adaptation input from PHY). Measurement configuration: RRC.

Solution to Exercise 2

SCTP is message-oriented (signalling messages keep their boundaries), reliable and congestion controlled like TCP, supports multiple streams so that a lost message of one UE does not block messages of others (no head-of-line blocking across streams), and supports multi-homing: an association can use several IP paths. If one transport link between eNB and MME fails, the association continues over the other address without tearing down the S1 interface and every UE context with it.

Solution to Exercise 3

$W \geq R\,T_{\text{RTT}}/(8L) = 10^9 \times 0.015/12\,000 = 1250$ PDUs. The SN space must be at least $2W = 2500$, so 12 bits (4096) are needed; LTE Release 8's 10 bits are insufficient (hence the longer RLC sequence numbers of later releases; NR RLC AM uses 12 or 18 bits).

Solution to Exercise 4

IPv6/UDP/RTP: $40 + 8 + 12 = 60$ bytes. Without ROHC: $61 + 60 = 121$ bytes per 20 ms = 48.4 kbit/s; with ROHC (3 bytes): 64 bytes = 25.6 kbit/s. Gain $121/64 = 1.89$. The gain is smaller than for 32-byte frames because the payload is larger.

Solution to Exercise 5

Each packet keeps the UE awake for about T_{in}; with $\lambda T_{\text{in}} \ll 1$ these periods rarely overlap. During the remaining time DRX applies. Fraction awake $f \approx \lambda T_{\text{in}} + (1 - \lambda T_{\text{in}})\,T_{\text{on}}/T_{\text{DRX}} = 0.02 + 0.98 \times 0.03125 = 0.0506$, i.e. 5.1%. With the powers of Section 6.4, $\bar{P} \approx 0.0506 \times 500 + 0.9494 \times 10 = 34.8\,\text{mW}$. The inactivity timer adds about 40% to the energy of the pure DRX case for this traffic.

Solution to Exercise

Each UE causes $1/30$ transitions per second, so $N = 10^4 \times 30 = 3 \times 10^5$ UEs per eNB – more than any cell carries, but the MME sees the sum over many eNBs, and paging, NAS and S1 messages load the core too. With RRC_INACTIVE a resumption needs only about 3 radio messages and no core signalling (the core sees the UE as connected), so the core load disappears and the radio load drops by about $4\times$, at the cost of storing contexts in the RAN.

13 Further reading

The E-UTRAN overall description [1] is the best entry point; the layer specifications are MAC [2], RLC [4], PDCP [3] and RRC [5]; the network interfaces are S1AP [6] and X2AP [7]; SCTP is specified in [10]. Sesia et al. [9] devote several chapters to the layer-2 protocols.

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